SURGITRACK – MASTER DEVELOPMENT PROMPT (FINAL VERSION)

Professional Surgical Training, Clinical Logbook & Medical Portfolio Platform

ROLE AND DEVELOPMENT OBJECTIVE

Act as a world-class software architect, senior full-stack developer, medical informatics engineer, database architect, and professional healthcare UI/UX designer.

Your task is to DESIGN AND BUILD a fully functional, cloud-based medical application named:

SurgiTrack – Professional Surgical Training & Clinical Portfolio

This application is intended for doctors, surgical trainees, clinical volunteers, and medical professionals who need to document their clinical cases, surgical exposure, operations, supervised training, skills, training hours, and professional achievements.

The main objective is to create a reliable, long-term professional portfolio that can be exported and presented to hospitals, academic institutions, medical licensing authorities, and professional organizations in Germany and internationally.

IMPORTANT DEVELOPMENT RULES:

- Build a real application, not a static design or prototype.
- Implement all forms, database operations, calculations, authentication, and exports.
- Do not create fake buttons or simulated functionality.
- Do not use hardcoded dashboard statistics.
- Do not remove existing working features when updating the application.
- Use a modular architecture and reusable components.
- Test every major feature before declaring it complete.
- If a service requires credentials, explain the configuration process instead of pretending it works.

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1. BRAND IDENTITY AND USER INTERFACE

Application name: SurgiTrack.

Tagline:

Document Every Case. Track Every Skill. Build Your Surgical Career.

Create an original, premium medical technology identity.

Logo and application icon

Design an original SVG logo combining:

- A simplified surgical instrument.
- A medical heartbeat line.
- A subtle professional shield or circular structure.

The logo must be suitable for:

- Website favicon.
- Android PWA icon.
- Desktop installation icon.
- Login screen.
- Application header.
- Loading screen.

Visual identity

Use this color palette:

- Deep Navy: #101C32
- Medical Teal: #0D9488
- Light Teal: #5EEAD4
- White: #FFFFFF
- Background: #F4F7FA
- Secondary Text: #64748B
- Success: #16A34A
- Warning: #D97706
- Error: #DC2626

Create a premium interface with:

- Modern typography.
- Professional cards.
- Rounded corners.

- Clean navigation.
- Interactive charts.
- Subtle animations.
- Consistent spacing.
- Responsive layouts.
- Light and dark themes.

Languages

Support:

1. English.

2. Arabic RTL.

3. German LTR.

Default language: English.

Provide a language selector in the profile and settings.

Medical terminology must be translated accurately.

The German version must use professional medical terminology suitable for a German hospital environment.

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2. TECHNOLOGY AND CLOUD INFRASTRUCTURE

Use a secure, scalable cloud architecture.

Preferred stack:

- Firebase Authentication.
- Cloud Firestore.
- Firebase Storage.
- Firebase Cloud Functions.
- Firebase Hosting.
- Firebase App Check.
- Gemini API through secure server-side functions.

Do not expose API keys or administrative credentials in frontend code.

Use environment configuration and server-side authorization.

Database persistence

All user information must be saved permanently in the cloud.

Closing the application, changing devices, or logging out must not delete saved records.

Implement real database persistence, not localStorage-only storage.

Use local storage only for appropriate temporary drafts, application preferences, and offline functionality.

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3. AUTHENTICATION AND ACCOUNT MANAGEMENT

Create professional login and registration interfaces.

Registration fields

- Full Name.
- Email.
- Password.
- Confirm Password.
- Country.
- Medical Qualification.
- Current Position.
- Specialty.
- Profile Photo (optional).

Required functionality

- Create Account.
- Login.
- Logout.
- Forgot Password.
- Password Reset Email.
- Email Verification.
- Persistent Authentication.

- Change Password.
- Change Email.
- Edit Profile.
- Change Profile Photo.
- Delete Account.

Authentication must use Firebase Authentication.

Requirements:

1. Reject duplicate email registrations.
2. Verify email addresses.
3. Reject incorrect email/password combinations.
4. Display clear authentication error messages.
5. Validate password strength.
6. Never store passwords in Firestore.
7. Restore the authenticated session after reopening the application.
8. Implement secure account deletion.

Each account must have a private portfolio.

Users must never access another user's clinical cases, operations, supervisors, documents, attachments, or private training information.

Implement Firestore and Storage Security Rules to enforce this.

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4. MAIN DASHBOARD

Create a professional dashboard named:

My Surgical Dashboard

Automatically calculate and display:

- Total Clinical Cases.
- Total Surgical Procedures.
- Total Training Sessions.
- Total Training Hours.
- Total Verified Training Hours, if an actual external verification record is entered.
- Total Unverified Training Hours.
- Total Supervisors.
- Total Hospitals.
- Total Skills Practiced.
- Total Assessed Competencies.
- Training Progress.

Create interactive charts:

1. Training Hours by Week.

2. Training Hours by Month.

3. Operations by Surgical Specialty.

4. Procedures by Hospital.

5. Procedures by Supervisor.

6. Surgical Participation Levels.

7. Skills Development.

8. Training Distribution.

Include:

- Recent Clinical Cases.
- Recent Surgical Operations.
- Recent Training Sessions.
- Recently Added Skills.
- Saved Surgical Articles.
- Quick Actions.

Every statistic must be calculated from actual database records.

Never insert demonstration figures into the real user dashboard.

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5. CLINICAL CASE LOGBOOK

Create a main navigation section:

Clinical Cases / My Logbook

Provide a prominent button:

+ Add New Clinical Case.

The form must be professional, comprehensive, and easy to complete on mobile.

5.1 Patient information

Fields:

- Automatically Generated Case Reference.
- Patient Age.
- Sex (optional).
- Presentation Date.
- Hospital.
- Department.
- Specialty.

STRICT PRIVACY RULE:

Never request or store:

- Patient Name.
- Patient ID.
- Hospital File Number.
- Address.
- Telephone Number.
- Facial photographs.
- Any direct patient identifier.

Use anonymous case reference numbers.

5.2 Clinical history

Create the following fields:

- Chief Complaint.
- History of Present Illness.
- Duration of Symptoms.
- Relevant Medical History.
- Surgical History.
- Drug History.
- Allergies.
- Family History (optional).
- Social History (optional).

5.3 Clinical examination

Expandable sections:

- General Examination.
- Vital Signs.
- Cardiovascular Examination.
- Respiratory Examination.
- Abdominal Examination.
- Neurological Examination.
- Local Examination.
- Other Relevant Findings.

Allow custom examination entries.

5.4 Investigations

Create an interactive searchable checklist.

Laboratory:

- CBC.
- Urea and Creatinine.
- Electrolytes.
- LFT.
- Coagulation Profile.
- CRP.

- ESR.
- Blood Glucose.
- HbA1c.
- Blood Group and Crossmatch.
- ABG.
- Other.

Imaging:

- X-ray.
- Ultrasound.
- Doppler Ultrasound.
- CT Scan.
- CT Angiography.
- MRI.
- MR Angiography.
- Echocardiography.
- Other.

Cardiac:

- ECG.
- Echocardiography.
- Cardiac Biomarkers.
- Other.

Allow multiple selections and custom investigations.

Each selected investigation can include:

- Date.
- Result.
- Summary.
- Optional attachment.

5.5 Diagnosis

Fields:

- Primary Diagnosis.
- Secondary Diagnoses.
- Differential Diagnosis.

- ICD-10 Code (optional).
- Clinical Assessment.

5.6 Management

Fields:

- Initial Management.
- Medical Management.
- Surgical Management.
- Treatment Plan.
- Clinical Outcome.
- Complications.

5.7 Surgical procedure

If applicable, display:

- Procedure Name.
- Procedure Date.
- Surgical Specialty.
- Indication.
- Surgical Approach.
- Operative Findings.
- Procedure Summary.
- Anaesthesia Type (optional).
- Outcome.

Participation selector:

- Observer.
- Assisted.
- First Assistant.
- Second Assistant.
- Performed Specific Steps Under Direct Supervision.
- Performed Under Supervision.
- Other.

Do not automatically classify observation as surgical assistance.

5.8 Follow-up

Optional fields:

- Follow-up Date.
- Clinical Progress.
- Wound Status.
- Complications.
- Additional Treatment.
- Final Outcome.

5.9 Clinical photographs and videos

Allow the user to upload:

- Clinical photographs.
- Operative photographs.
- Photographs of surgical wounds.
- Educational surgical videos.
- Supporting documents.

IMPORTANT PRIVACY AND CONSENT REQUIREMENTS:

The application must not store patient names, faces, identification numbers, or identifying information.

Clinical photographs may only be uploaded after appropriate patient consent has been obtained and institutional permissions are satisfied.

Before every clinical media upload, show a privacy confirmation:

"I confirm that this media is authorized for educational documentation, appropriate consent has been obtained, and no patient-identifying information is included."

Provide:

- Upload Progress.
- Image Preview.
- Delete Attachment.
- Replace Attachment.
- Secure Storage.

Implement private Firebase Storage permissions.

Do not automatically upload clinical photographs to external AI services.

5.10 Save and display

After saving:

- Validate fields.
- Store the case in Firestore.
- Upload authorized attachments.
- Generate a unique case ID.
- Update dashboard statistics.
- Show a successful save notification.
- Return to the case list.

Display cases as professional cards.

Allow:

- View.
- Edit.
- Delete.
- Search.
- Filter.
- Sort.
- Export.

The detailed case page must display the information chronologically from presentation to follow-up.

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6. SUPERVISOR MANAGEMENT

Create a separate main navigation icon:

My Training / Training Portfolio

The central organizational principle is:

Every supervisor has an independent profile, and every training activity recorded under that supervisor belongs to that supervisor's portfolio.

6.1 Add Supervisor

Button:

+ Add New Supervisor.

Fields:

- Full Name.
- Academic Degree.
- Academic Title.
- Professional Position.
- Specialty.
- Subspecialty.
- Department.
- Hospital.
- University (optional).
- Country.
- Email (optional).
- Additional Notes.

Examples of positions:

- Consultant Surgeon.
- Professor of Surgery.
- Associate Professor.
- Specialist Surgeon.
- Senior Registrar.
- Attending Surgeon.

Keep academic qualifications and professional positions as separate fields.

After saving, create an independent supervisor dashboard.

Example:

Supervisor A:

- Profile.
- Hospital.
- Training Period.
- Training Records.
- Operations.

- Skills.
- Statistics.
- Export Portfolio.

Supervisor B:

- Separate profile.
- Separate training records.
- Separate statistics.
- Separate exports.

Never mix the training records of two supervisors.

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7. SUPERVISED TRAINING AND SURGICAL OPERATIONS LOGBOOK

Inside every supervisor profile, provide:

+ Add Training Record

Fields should automatically retrieve supervisor and trainee information.

7.1 Supervisor details

Automatically populate:

- Supervisor Name.
- Academic Degree.
- Professional Position.
- Specialty.
- Department.
- Hospital.

7.2 Trainee details

Automatically populate:

- Full Name.
- Medical Qualification.
- Current Position.

- Specialty.
- Email.

Allow editing of the trainee's professional position.

Examples:

- Clinical Volunteer.
- Surgical Trainee.
- Junior Doctor.
- Medical Officer.
- Resident.

7.3 Training activity

Fields:

- Training Date.
- Hospital.
- Department.
- Specialty.
- Activity Type.
- Procedure Name.
- Linked Clinical Case Reference (optional).
- Start Time.
- End Time.
- Break Duration.
- Net Training Hours.

Calculate automatically:

Net Training Hours = End Time − Start Time − Break Duration.

Validate all times.

Prevent negative durations and duplicate session records.

7.4 Surgical exposure

Options:

- Observed Procedure.

- Assisted During Procedure.
- First Assistant.
- Second Assistant.
- Scrubbed and Assisted.
- Performed Specific Steps Under Direct Supervision.
- Performed Procedure Under Direct Supervision.
- Other.

7.5 Participation details

Fields:

- Description of Participation.
- Responsibilities.
- Surgical Techniques Observed.
- Surgical Techniques Practiced.
- Level of Supervision.
- Skills Acquired.
- Learning Points.
- Additional Notes.

7.6 Skills

Create a searchable skills checklist.

General Surgery:

- Surgical Hand Scrubbing.
- Sterile Gowning and Gloving.
- Surgical Instrument Handling.
- Suturing.
- Knot Tying.
- Skin Closure.
- Wound Management.
- Incision and Drainage.
- Surgical Exposure.
- Laparotomy Assistance.

Vascular Surgery:

- Peripheral Vascular Examination.
- Arterial Pulse Assessment.

- Doppler Assessment.
- Vascular Access.
- Vessel Exposure.
- Arterial Dissection.
- Vascular Clamping.
- Arteriotomy.
- Vascular Suturing.
- Arterial Anastomosis.
- Vein Harvesting.
- Bypass Surgery Exposure.
- Embolectomy Exposure.
- Peripheral Angiography Exposure.
- Endovascular Procedure Exposure.

Allow custom skills.

For each skill record:

- Skill Name.
- Date.
- Exposure Level.
- Number of Recorded Sessions.
- Duration.
- Supervision Level.
- Assessment.
- Supervisor Feedback.

7.7 Save

After saving:

- Link the record to the selected supervisor.
- Update the supervisor's dashboard.
- Update the overall user dashboard.
- Recalculate hours.
- Update procedure counts.
- Update skill statistics.
- Add the record to chronological history.

Provide:

+ Add Another Training Record.

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8. TRAINING SESSIONS – PRECISE HOURS TRACKING

Create a dedicated Training Sessions module.

This module is essential because training hours must be calculated from real documented attendance.

Allow the user to create a training session independently or associate it with a surgical procedure.

Fields:

- Supervisor.
- Hospital.
- Department.
- Training Date.
- Session Start Time.
- Session End Time.
- Break Duration.
- Session Type.
- Training Activities.
- Skills Practiced.
- Optional Notes.

Automatically calculate net hours.

Support multiple sessions on the same day when appropriate.

Weekly tracking

Calculate:

- Daily Hours.
- Weekly Hours.
- Monthly Hours.
- Average Weekly Hours.
- Training Days.
- Total Hours per Supervisor.
- Total Hours per Hospital.

Important calculation rules

1. Do not infer training hours from the number of operations.
2. Do not automatically assign a standard duration to an operation.
3. Subtract recorded breaks.
4. Prevent duplicate sessions.
5. Distinguish actual attendance from estimated time.
6. Keep a record of edited session times.
7. Do not count overlapping sessions twice in aggregate attendance hours.

Provide clear weekly and monthly charts.

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9. SKILL MATRIX AND COMPETENCY PROGRESS

Create a separate section:

My Skills & Competencies

This section must maintain an independent cumulative record of professional skills.

Skill Matrix

Organize skills by specialty and category.

For each skill, display:

- Skill Name.
- Total Recorded Sessions.
- Total Documented Hours.
- Dates Practiced.
- Supervisors.
- Hospitals.

- Exposure Level.
- Assessment Status.
- Competency Progress.

Exposure levels

Use a clear distinction:

1. Observed.

2. Assisted.

3. Performed Specific Steps Under Direct Supervision.

4. Performed Under Direct Supervision.

5. Assessed by Supervisor.

Competency status

Use:

- Not Yet Assessed.
- Learning.
- Practicing.
- Supervisor Assessed.
- Competency Confirmed.

Only assign confirmed competency when an actual assessment has been entered.

Never automatically convert repeated exposure into confirmed competency.

Progress visualization

Provide:

- Skill Progress Bars.
- Skill Distribution.
- Skills by Supervisor.
- Skills by Specialty.
- Skills by Hospital.
- Timeline of Skill Development.

Allow exporting the Skill Matrix into Excel and PDF.

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10. TRAINING ANALYTICS AND PROGRESS DASHBOARD

Every supervisor must have an independent analytics page.

Calculate:

- Total Cases.
- Total Surgical Procedures.
- Total Training Sessions.
- Total Documented Hours.
- Total Weekly Hours.
- Total Monthly Hours.
- Training Start Date.
- Last Training Date.
- Calendar Training Duration.
- Active Training Days.
- Average Weekly Hours.
- Skills Practiced.
- Competencies Assessed.
- Participation Levels.

Display:

1. Training Hours by Week.

2. Operations by Type.

3. Hours by Skill.

4. Training Activity Timeline.

5. Participation Distribution.

6. Clinical Exposure by Category.

Automatically calculate the training period using the earliest and latest recorded training dates for that supervisor.

Clearly distinguish calendar duration from actual attendance hours.

The overall dashboard must aggregate all supervisors without mixing their individual records.

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11. SUPERVISOR-SPECIFIC EXPORT SYSTEM

This is a critical module.

Inside each supervisor profile, add:

Export Training Portfolio

The user must be able to export each supervisor's complete training period independently.

Formats:

- Excel (.xlsx).
- Word (.docx).
- PDF (.pdf).

Excel export – mandatory

Create a professionally formatted workbook.

Include these columns:

1. Trainee Name.
2. Supervisor Name.
3. Academic Degree.
4. Professional Position.
5. Specialty.
6. Department.
7. Hospital.
8. Training Date.
9. Procedure Date.
10. Procedure Name.
11. Clinical Case Reference.
12. Training Activity.
13. Surgical Exposure.
14. Participation Role.
15. Responsibilities.
16. Skills Acquired.
17. Start Time.
18. End Time.
19. Break Duration.
20. Net Training Hours.
21. Supervision Level.
22. Supervisor Feedback.
23. Additional Notes.

Excel workbook structure

Sheet 1: Training Summary.

Sheet 2: Detailed Training Records.

Sheet 3: Surgical Procedures.

Sheet 4: Skills Matrix.

Sheet 5: Weekly Hours.

Summary sheet

Automatically display:

- Trainee Name.
- Supervisor Name.
- Hospital.
- Department.
- Training Period.
- Total Cases.
- Total Operations.
- Total Sessions.
- Total Hours.
- Hours per Week.
- Skills Practiced.
- Participation Distribution.

Official signature and stamp area

At the end of the exported workbook, create a printable certification section containing:

Supervisor Full Name.

Academic Degree.

Professional Position.

Department.

Hospital.

University (if applicable).

Statement confirming that the recorded training activities have been reviewed.

Date.

Official Signature.

Official Hospital / Department Stamp.

Leave the signature and stamp areas blank.

IMPORTANT:

The application must NOT generate a fake signature or official stamp.

The user will download and print the Excel document, present it to the supervisor, and obtain the official signature and stamp manually.

Do not create an in-app signature approval or supervisor submission system.

The export must be professionally designed, easy to print, and suitable for official review.

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12. RECOMMENDATION LETTER GENERATOR

Create a separate navigation section:

Recommendation Letters

12.1 Select supervisor

Provide a dropdown containing all supervisors registered by the user.

When selecting a supervisor, automatically retrieve the corresponding data.

Supervisor:

- Name.
- Academic Degree.
- Professional Position.
- Specialty.
- Department.
- Hospital.
- University.

Trainee:

- Full Name.
- Position.
- Qualification.
- Specialty.
- Email.

Training:

- Start Date.
- Last Training Date.
- Calendar Duration.
- Total Training Hours.
- Average Weekly Hours.
- Total Clinical Cases.
- Total Surgical Procedures.
- Total Training Sessions.
- Skills Practiced.

12.2 Clinical duties
Automatically retrieve documented activities, such as:
Patient Assessment.
Clinical History Taking.
Physical Examination.
Clinical Case Discussion.
Preoperative Assessment.
Postoperative Follow-up.
Ward Rounds.
Surgical Assistance.
Other Recorded Activities.
12.3 Skills
Automatically retrieve all skills recorded under the selected supervisor.
Only include information that exists in the database.

12.4 AI recommendation drafting

Use Gemini to draft a formal professional recommendation letter in:

English.

German.

Arabic.

The letter must have:

Professional heading.

Date.

Recipient.

Supervisor introduction.

Trainee introduction.

Training relationship.

Training period.

Clinical duties.

Surgical exposure.

Documented hours.

Number of cases and procedures.

Skills acquired.

Professional evaluation.

Recommendation statement.

Supervisor signature area.

The AI must never invent:

Supervisor opinions.

Clinical achievements.

Competencies.

Training hours.

Operations.

Qualifications.

Endorsements.

The supervisor's personal assessment must be supplied or approved by the supervisor.

The user must be able to edit the draft before exporting it.

12.5 Recommendation design

Create a premium academic medical document.

Use:

White background.

Professional letterhead.

Elegant typography.

Clear heading hierarchy.

Proper margins.

Highlighted trainee name.

Structured paragraphs.

Supervisor details at the bottom.

Blank signature and official stamp areas.

12.6 Export

Provide:

Word (.docx).

PDF.

Print.

Edit.

Regenerate Draft.

The generated document is a draft for the supervisor to review, finalize, sign, and stamp outside the application.

13. GERMAN MEDICAL PORTFOLIO EXPORT

Create a dedicated German export mode.

This is particularly important because the user intends to use the portfolio for professional recognition and employment in Germany.

Provide:

Export in German – Deutsches Tätigkeits- und Operationsportfolio

Use accurate professional German terminology.

Examples:

Klinische Tätigkeit.

Chirurgische Weiterbildung.

Operative Erfahrung.

Assistenz bei operativen Eingriffen.

Teilnahme an chirurgischen Eingriffen.

Unter direkter Supervision durchgeführte operative Teilschritte.

Dokumentierte Weiterbildungsstunden.

Betreuender Facharzt.

Klinik / Krankenhaus.

Abteilung für Gefäßchirurgie.

Klinische Kompetenzen.

Operative Tätigkeiten.

Weiterbildungszeitraum.

The system must distinguish accurately between:

Beobachtet.

Assistiert.

Als Erstassistent tätig.

Als Zweitassistent tätig.

Einzelne operative Schritte unter direkter Supervision durchgeführt.

Do not translate every participation type into the same German expression.

German export documents

Generate:

Deutsches Operationslogbuch.

Deutsches Weiterbildungsportfolio.

Übersicht der klinischen Tätigkeiten.

Übersicht der dokumentierten Weiterbildungsstunden.

Kompetenzmatrix.

Zusammenfassung der chirurgischen Erfahrung.

Provide Word, Excel, and PDF export.

Use formal medical German and professional document formatting.

Do not claim that the exported document automatically meets every German authority's specific certification requirements.

14. BACKUP AND RESTORE SYSTEM

Create a dedicated section:

Backup & Restore

The user must be able to protect their entire portfolio.

Backup

Provide:

Create Full Backup.

Download Backup.

Backup Date.

Backup Size.

Backup Status.

Last Successful Backup.

The backup should include:

Clinical Cases.

Training Records.

Supervisors.

Training Sessions.

Skills.

Recommendations.

User Preferences.

Permitted Attachments and their references.

Use a structured archive format with JSON data and associated files, such as ZIP.

Provide clear export and backup status.

Restore

Allow the user to:

Select Backup File.

Preview Backup Contents.

Validate Backup Integrity.

Restore Data.

Confirm Restoration.

Implement safeguards against accidental duplication.

Use unique record identifiers and restore operations that detect existing records.

Never silently overwrite existing records.

Explain the difference between:

A user-downloaded backup.

Firebase server-side backups.

A data export.

Where Firebase scheduled backup services require billing or additional configuration, clearly explain the requirement.

15. OFFLINE DRAFTS AND AUTOMATIC SYNCHRONIZATION

This is a mandatory feature.

The application must remain useful when the user has no internet connection.

Implement a reliable offline-first workflow.

15.1 Offline data entry

Allow the user to:

Create a clinical case draft.

Create a training record draft.

Create a training session.

Record skills.

Edit previously downloaded records where supported.

Use Firestore offline persistence and an appropriate local draft mechanism.

15.2 Synchronization

When internet connection returns:

Automatically synchronize pending records.

Upload queued attachments where authorized.

Update database records.

Refresh dashboard statistics.

Prevent duplicate entries.

Display synchronization status.

15.3 Sync indicators

Display:

Online.

Offline.

Pending Synchronization.

Synchronizing.

Successfully Synchronized.

Synchronization Error.

15.4 Conflict handling

If the same record has been edited on multiple devices:

Detect conflicts.

Preserve the original data.

Show the conflicting versions.
Allow the user to choose which version to retain.
Do not silently overwrite important clinical or training records.
Use unique IDs and idempotent synchronization operations.
Do not count a pending offline record as a successfully synchronized cloud record.
Clearly distinguish local draft data from saved cloud data.

16. SURGICAL NEWS AND INTERNATIONAL MEDICAL RESOURCES

Create a separate section:
Surgical News & Updates
Connect to legitimate, accessible medical information sources.
Suggested sources:
International Organizations:
Royal College of Surgeons of England.
American College of Surgeons.
Society for Vascular Surgery.
European Society for Vascular Surgery.
World Health Organization.
Journals and Research:
British Journal of Surgery.
Annals of Surgery.
Journal of Vascular Surgery.
European Journal of Vascular and Endovascular Surgery.
BMJ.
The Lancet.
JAMA Surgery.
PubMed.
News categories
General Surgery.
Vascular Surgery.
Cardiothoracic Surgery.
Neurosurgery.
Pediatric Surgery.
Trauma Surgery.
Surgical Oncology.
Transplant Surgery.
Surgical Education.
Medical Research.
Automatic updates
Use:
Official RSS feeds.

Authorized APIs.

Publicly accessible feeds.

Do not assume every journal has an accessible API.

Do not scrape restricted websites or bypass access controls.

If a source has no legitimate feed, provide an external source link.

Use secure server-side scheduled synchronization.

Display:

Article Title.

Publication Date.

Publisher.

Category.

Short Summary.

Original Article Link.

Last Synchronization Date.

Provide:

Search.

Filter.

Bookmark.

Save Article.

Open Original Article.

Do not reproduce full copyrighted articles.

Create an administrator interface to manage news sources and synchronization status.

17. AI MEDICAL ASSISTANT

Create an optional Gemini-powered assistant.

It can help with:

Medical English.

German medical terminology.

Translation of generic professional text.

Surgical terminology.

Structuring blank clinical documentation templates.

Improving grammar.

General educational explanations.

Preparing non-patient-specific portfolio summaries.

STRICT DATA PROTECTION:

Do not transmit clinical case records, patient-related narratives, photographs, or videos to Gemini or any other external AI provider.

The AI assistant must work with general medical educational content or explicitly non-clinical, non-identifying information.

Do not automatically process the contents of the clinical case database.

Do not generate fabricated clinical findings, diagnoses, procedures, training hours, or supervisor assessments.

All AI requests must use secure server-side functions.

18. USER PROFILE AND SETTINGS

Create a dedicated profile page.

Fields:

Profile Picture.

Full Name.

Email.

Country.

Medical Qualification.

Position.

Specialty.

Hospitals.

Professional Biography.

Additional Qualifications.

MRCS or other professional certificates.

Allow:

Edit Profile.

Change Photo.

Change Password.

Change Email.

Language.

Theme.

Notification Settings.

Privacy Settings.

Download My Data.

Backup.

Restore.

Logout.

Delete Account.

19. ADMINISTRATION PANEL

Create a protected administrator dashboard.

Roles:

Super Administrator.

Normal User.

Implement roles securely using trusted backend permissions and Firestore Security Rules.

User Management

Administrator can:

View registered accounts.

Search users.

Filter users.

View basic account information.

View registration date.

Suspend accounts.

Reactivate accounts.

Delete accounts.

Manage roles.

Administrators must not automatically gain access to private clinical cases, medical photographs, or private training notes.

Application Management

Administrator can:

Manage application sections.

Add or remove navigation items.

Edit descriptions.

Change icons.

Manage default skills.

Manage specialties.

Manage investigation checklists.

Manage news sources.

Configure application settings.

Audit Log

Record administrative actions:

Administrator ID.

Action.

Timestamp.

Affected Record.

Reason where applicable.

20. DATABASE ARCHITECTURE

Use a scalable Firestore architecture.

Suggested structure:

users/{userId}

users/{userId}/clinicalCases/{caseId}

users/{userId}/supervisors/{supervisorId}

users/{userId}/supervisors/{supervisorId}/trainingRecords/{recordId}

users/{userId}/trainingSessions/{sessionId}

users/{userId}/skills/{skillId}

users/{userId}/recommendations/{recommendationId}

users/{userId}/savedArticles/{articleId}

users/{userId}/notifications/{notificationId}

users/{userId}/exports/{exportId}

users/{userId}/backups/{backupId}

public/surgicalNews/{articleId}

public/newsSources/{sourceId}

system/settings
system/auditLogs
You may optimize this architecture where necessary.
Maintain strict user ownership and clear relationships between:
Clinical Cases.
Surgical Procedures.
Supervisors.
Hospitals.
Training Sessions.
Skills.
Training Records.
Recommendations.
Use appropriate Firestore indexes, timestamps, validation, pagination, and secure deletion.

21. EXPORT ENGINE

Implement real export generation.
Required formats:
Excel XLSX.
Word DOCX.
PDF.
Exports must work on mobile and desktop.
Use appropriate document-generation libraries.
Preserve:
Medical terminology.
Arabic and German Unicode text.
Tables.
Dates.
Numbers.
Training hours.
Supervisor information.
Professional formatting.
Test that every generated file opens correctly.
Never show a download button that does not produce an actual file.

22. SECURITY AND PRIVACY

This application may contain sensitive clinical and professional information.
Implement privacy by design.
Mandatory safeguards:
No patient names.
No patient identification numbers.
No patient faces.
No unauthorized clinical photographs.

Consent confirmation before media upload.

Private Firebase Storage.

Secure authentication.

Firestore Security Rules.

Storage Security Rules.

Server-side role verification.

Secure API key management.

File type validation.

File size restrictions.

Account deletion.

User-controlled backup and restore.

Audit logging for administrative actions.

Do not claim GDPR, HIPAA, or other formal regulatory compliance without the required legal and technical verification.

23. DEVELOPMENT PHASES

Implement the application in the following order.

Phase 1 – Foundation

Logo.

UI system.

Authentication.

Firebase connection.

User profile.

Dashboard.

Phase 2 – Clinical Logbook

Add cases.

Clinical history.

Examination.

Investigations.

Diagnosis.

Management.

Procedures.

Follow-up.

Secure attachments.

Phase 3 – Supervisors and Training

Supervisor registration.

Independent supervisor profiles.

Training records.

Surgical exposure.

Participation roles.

Hospital records.

Phase 4 – Training Sessions and Skills

Session tracking.

Hours calculation.

Skill Matrix.

Competency Progress.

Weekly analytics.

Monthly analytics.

Phase 5 – Portfolio Exports

Supervisor-specific Excel.

Word exports.

PDF exports.

Signature and stamp areas.

German Portfolio Export.

Phase 6 – Recommendation Generator

Supervisor selection.

Automatic data retrieval.

Gemini letter drafting.

Editing.

Word and PDF exports.

Phase 7 – Backup and Offline

Offline drafts.

Automatic synchronization.

Conflict handling.

Backup.

Restore.

Phase 8 – Surgical News

RSS.

Authorized APIs.

Automatic synchronization.

News administration.

Phase 9 – Admin and Security

User management.

Roles.

Security Rules.

Audit logs.

Account deletion.

Final testing.

24. TESTING AND QUALITY ASSURANCE

Before declaring the application complete, test these scenarios.

Authentication

Create an account.

Verify email.
Log in.
Log out.
Reject incorrect password.
Restore session.
Clinical Logbook
Add a case.
Edit a case.
Delete a case.
Attach authorized media.
Export a case.
Supervisors
Add two supervisors.
Add multiple operations under each.
Confirm complete separation of records.
Edit and delete records.
Training
Create training sessions.
Calculate hours.
Subtract breaks.
Calculate weekly totals.
Detect overlapping sessions.
Prevent duplicate records.
Skills
Record a skill.
Update exposure level.
Record supervisor assessment.
Verify that exposure does not automatically equal competency.

Export
Generate Excel.
Open Excel.
Generate Word.
Open Word.
Generate PDF.
Open PDF.
Confirm supervisor-specific data separation.
Confirm signature areas are blank.
German Portfolio
Generate German reports.
Verify professional terminology.

Verify dates and hours.
Verify Unicode formatting.
Backup and Offline
Create a backup.
Restore a backup.
Create offline records.
Reconnect.
Confirm synchronization.
Confirm no duplicates.
Test conflict handling.
Security
Test user data isolation.
Test Storage permissions.
Test administrator privileges.
Test account deletion.
Final deployment
Test mobile responsiveness.
Test desktop.
Test PWA installation.
Test persistence after closing the app.

25. STRICT EXECUTION INSTRUCTIONS FOR GEMINI

Do not interpret this document as a request to merely explain how the application could be built.
You must implement the actual application in the connected Google AI Studio project.
Follow these instructions:
First inspect the available project files and existing architecture.
Determine the existing Firebase configuration.
Preserve working infrastructure and functionality.
Create a clear implementation plan.
Build actual application components.
Connect all forms to real database operations.
Implement secure backend functions.
Implement calculations and export generation.
Implement the responsive user interface.
Test the features after implementation.
Do not create a homepage and claim the project is complete.
Do not replace working Firebase authentication with simulated authentication.
Do not create fake user accounts.
Do not use static numbers in analytics.
Do not use fake Excel, Word, or PDF downloads.
Do not use fabricated news items.
Do not generate fake supervisor approvals.

Do not create frontend-only administrator permissions.

Do not silently remove any previously working functionality.

Implementation reporting

After each development phase, report:

What was implemented.

Which database collections were created.

Which Firebase services were connected.

Which security rules were added.

Which features were tested.

Which features still need configuration.

Which errors remain.

If a requested feature cannot be implemented with the available infrastructure, explain the exact technical limitation and provide the closest legitimate implementation.

FINAL OBJECTIVE

Deliver a real, secure, persistent, cloud-based application called SurgiTrack.

The finished platform must enable a medical professional to:

Record clinical cases.

Document surgical operations.

Track participation levels.

Register multiple supervisors.

Record training sessions.

Calculate accurate training hours.

Track surgical skills.

Monitor competency development.

Generate supervisor-specific training portfolios.

Export official-review-ready Excel documents.

Generate professional recommendation letter drafts.

Export medical portfolios in German and English.

Back up and restore records.

Work offline and synchronize automatically.

Follow international surgical news.

Prioritize accuracy, professional documentation, data security, usability, and real functionality.

Build the application as a long-term professional medical portfolio system, not merely a simple medical diary.